WORKSHEET · SCRATCH (GR 6–8) INTERMEDIATE: DATA & SIMULATIONS

Lesson 1: Lists and Traversal

Module 1 | 45 minutes | Lists and traversal | Grades 6–8 · ages 11–14

Name: _____ Class: _____ Date: _____

Practise what you learned today. Read each task carefully.

1. Key words

Match each word to what it means. Write the letter.

a. traverse ___ A. A loop that starts or stops one step wrong.
b. index ___ B. A variable that builds a result in a loop.
c. accumulator ___ C. The position of an item (from 1 in Scratch).
d. off-by-one error ___ D. Visit every item in a list.

2. Plan your code

Before you build it: Total the week: temps holds 7 temperatures. Use an index i and a total accumulator to add up every item, then say the total (it should be 153). Write or draw the blocks you will use.

3. Off by one

A loop starts with i = 0 and reads item i of a 5-item list each time. What goes wrong? Circle one.

a. A. Item 0 doesn't exist, so the first read is empty and the last item is missed
b. B. Nothing — lists start at 0
c. C. It reads every item twice

4. Plan your code

Before you build it: Hot days: count how many temps are above 20. Traverse the list; if item i > 20, change hot by 1. Say the count (it should be 4). Write or draw the blocks you will use.

5. Trace in Python

What does this Python print? Code: scores = [5, 9, 2] ↵ count = 0 ↵ for s in scores: ↵ if s > 4: ↵ count = count + 1 ↵ print(count)

a. A. 2
b. B. 3
c. C. 16

How did it go? Colour a face: 😀 I can do it · 🙂 Nearly there · 🤔 I need help

☐ I can traverse a list with an index.
☐ I can accumulate a total or count.
☐ I can spot off-by-one errors.